

## **Statement of Previous Research**

### **Characterization of Internal Flows Through Transitioning Ducts with Globally Favorable and Adverse Pressure Gradients**

Vincent Onoja, PhD Candidate (Defense date: Aug. 26, 2026)

Over the past five years, my research has centered on the experimental fluid mechanics of internal flows within shape-transitioning ducts. These cross-sectional-area-changing ducts with streamline curvature are at the heart of compact, highly-integrated propulsion systems such as blended-wing-body embedded-engine inlets and non-axisymmetric nozzles. This work was carried out at the University of Cincinnati's Advanced Laboratory for Fluids and Acoustics (ALFA), under Dr. Daniel Cuppoletti, in collaboration with the Massachusetts Institute of Technology and the U.S. Naval Air Warfare Center Aircraft Division (NAWCAD), and sponsored by the Office of Naval Research (ONR) and the Air Force Office of Scientific Research (AFOSR). It culminated in my doctoral dissertation and in a predictive scaling theory for the streamwise vortices these geometries generate.

This program grew out of an earlier proof-of-concept study, recently accepted for publication in MDPI Aerospace, in which I characterized the internal flow of two mildly shape-transitioning nozzle ducts using planar PIV, which was used to validate  $k-\omega$  SST RANS simulations. That study established the measurement-and-validation workflow on tractable geometries before it was extended to the aggressive asymmetric ducts of the funded program: it demonstrated that PIV and RANS agreed to within roughly 1–2% in the upstream and mid-duct regions, identified where steady RANS begins to lose fidelity near the exit, and validated the discharge-coefficient and boundary-layer measurements against independent pitot probe

measurements. Critically, it also revealed the mechanism that would become central to my dissertation: the diverging sidewall introduces a local adverse pressure gradient that thickens the boundary layer, the same wall-driven forcing that I later identified as the initiator of streamwise-vortex generation in the asymmetrically shape transitioning ducts. The benign-duct study thus both de-risked the experimental approach and surfaced the physics that motivated the duct-asymmetry investigations that followed.

In 2021, I designed and built a cost-effective test facility for investigating aggressively shaped ducts. The facility accommodates interchangeable, mathematically-defined duct configurations with walls that are specified by piecewise-polynomial definitions, allowing the local wall shape to be varied methodically, while global parameters (contraction ratio, exit aspect ratio, and transition length) are held fixed, so that the effect of wall shaping can be isolated from bulk area change. I characterized these geometries using high-speed planar and stereoscopic particle image velocimetry, pitot-static probes, and unsteady Kulite pressure measurements. I complemented these experiments by developing Python-based feature-detection and vortex-tracking algorithms that automatically identified coherent structures and quantified circulation and vortex-core trajectories from noisy velocity fields. I also performed detailed boundary-layer analyses to quantify differences in growth between the concave and convex walls throughout the shape transition.

My work proceeded along three connected areas:

First, I conducted a multi-fidelity characterization and turbulence-model assessment of six shape-transitioning duct configurations. I benchmarked high-resolution experimental measurements against a simulation suite spanning my own  $k-\omega$  SST Reynolds-averaged Navier-Stokes computations in ANSYS FLUENT, the Spalart-Allmaras model family and its solution-

adaptive variants implemented by collaborators at MIT, and wall-modeled large-eddy simulations performed with the CharLES solver at NAWCAD. The comparison quantified the predictive fidelity of each approach. First-order quantities, including the mean velocity field and pressure recovery, were reproduced well by all models. However, differences emerged in the prediction of second-order flow features like spanwise and streamwise vorticities, as well as in-plane Reynolds stresses. The RANS models consistently underpredicted the primary vortex circulation and misplaced the vortex core relative to the experiments, whereas the wall-modeled LES accurately captured both the circulation and vortex structure. These results established the fidelity required to reliably predict secondary-flow structure in this class of ducts.

Second, and as the central contribution of my dissertation, I identified a geometry-vortex scaling law for a class of contracting shape-transitioning ducts whose converging and diverging walls are independently prescribed by distinct geometric contours. Using circulation measurements at multiple streamwise stations, the corresponding vorticity-weighted centroid trajectories, and dimensional analysis, I found that the exit-plane vortex circulation is governed primarily by the first moment of the wall-slope-asymmetry distribution: the streamwise location at which the asymmetry between the converging and diverging walls is concentrated.

Physically, upstream-loaded asymmetry causes the diverging wall to impose its adverse pressure gradient earlier in the transition, allowing the resulting vortex a longer streamwise distance over which to develop and intensify, whereas downstream-loaded asymmetry produces a weaker vortex. Furthermore, the downstream decay through the straight section was found to be nearly universal, largely independent of the upstream wall profile.

This work establishes the first geometry-based predictive scaling for streamwise vortices generated by asymmetric wall shaping in contracting shape-transitioning ducts. Rather than relying

on global geometric measures or centerline-curvature descriptors developed for serpentine ducts, the scaling directly relates the spatial distribution of wall geometry to vortex strength in a class of flows driven by asymmetric wall transitions.

Third, I advanced the experimental methodology itself. Measuring three-component velocity and vorticity inside confined, optically difficult transitioning geometries required techniques for reflection mitigation, and uncertainty quantification that few laboratories have developed, together with the automated structure-extraction methods that make such measurements usable as quantitative validation data.

### **Outcomes and impact.**

The benchmark datasets I produced have been adopted beyond my own group and are now used for solver and turbulence-model validation by collaborators at MIT, directly informing computational method development in the field. These results have been disseminated through first-author publications and presentations, including AIAA SciTech 2023, the AIAA AVIATION Forum 2025, and ISPIV 2025 (Tokyo), with additional journal articles currently under peer review. I also serve on the AIAA Inlets, Nozzles, and Propulsion System Integration Technical Committee and have engaged with ONR through program reviews and quarterly reporting, working closely with NAWCAD engineers in a collaborative research context. More broadly, this work advances fundamental understanding of secondary flows in asymmetric internal geometries and establishes validation datasets that clarify fidelity requirements for their prediction. Both the scaling framework and the supporting datasets directly extend into the generalization program I propose to pursue as a Distinguished Fellow.

### **Publications can be found here:**

Google Scholar Profile:

<https://scholar.google.com/citations?user=Mjds1ncAAAAJ&hl=en&oi=ao>